

Engineering Change Notice(ECN)

ECN-D-2603-01(Requirement for TP-Host key switch)

FA연구소 설계2팀

작성	검토	승인
임제우 프로	최동수 팀장	이태경 본부장
2/13	2/13	2/13

NINEBELL

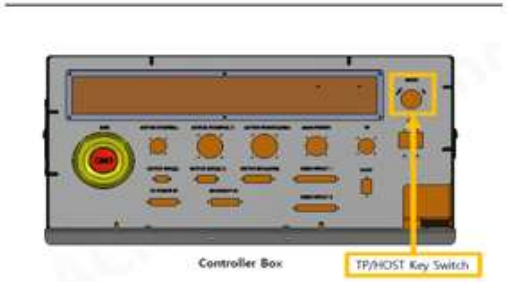
1. Engineering Change Request

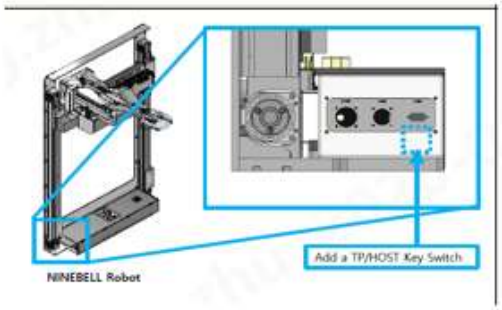
NINEBELL	설계 변경 요청서(ECR) (Engineering Change Request)		Customer	AOI
			ECR No.	

Requesting department	Name / Department	/	Change ITEM	☐ Customer Repair ☐ Cost reduction ☐ Quality improve ☐ Work Efficiency ☐ Parts discontinued ☐ ETC
	Date	2020-02-10		

Request project	Category	Project Type	변경 level (Change Level)	☑ Level 1	Design structure changes are required.
	☐ NP ☐ ES			☐ Level 2	Simple part and work method change

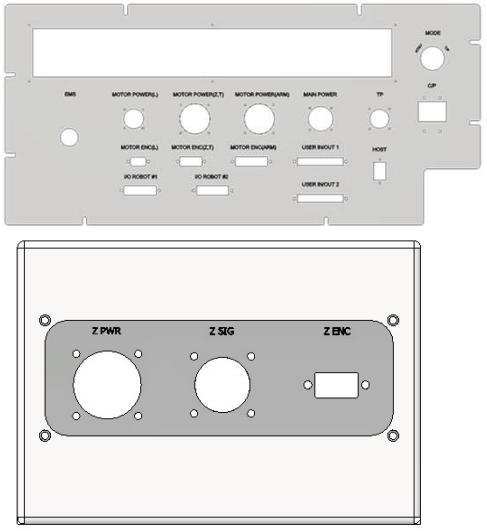
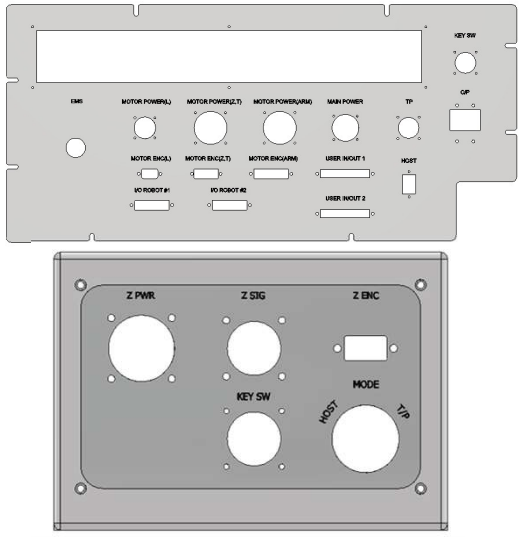
Before(Problems and Change Request)	After(Expected Effect)
将TP host的按钮更换位置，便于调试人员操作	



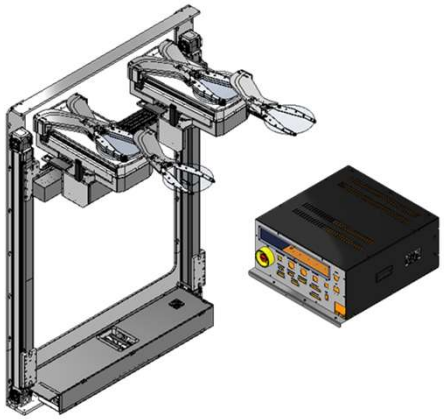
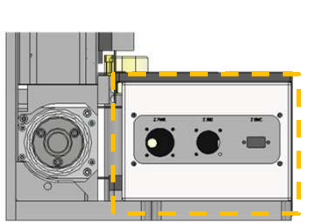
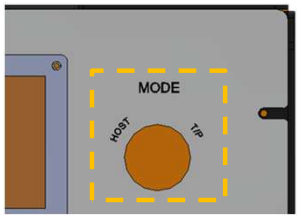
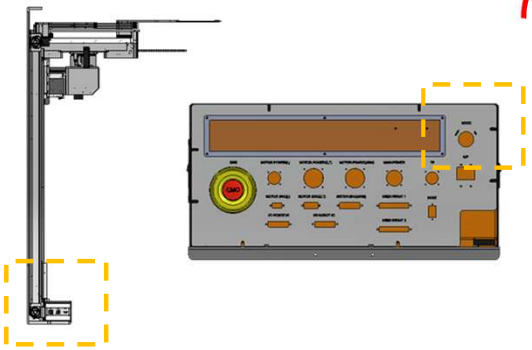


Related Division	☐ Mechanical design	Potential issue	☐ Instrument & tool additional cost incurred
	☑ Electrical design		☐ Work process change and work time
	☐ Manufacturing		☐ Mechanical failure
	☐ Purchasing & Materials		☐ ETC
	☐ Quality		

2. Engineering Change Notice

적용 장비 타입 Applicable project type	All Single Robot	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	■ 고객 요청 (Customer request) □ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) □ 기타 (ETC)	ECR NO.	ECR-S-2603-01
		ECN NO.	ECN-D-2603-01
변경 상세 사유 Detailed reason for change	컨트롤박스에 설치된 TP-HOST 스위치를 조작하기 위해서는 작업자가 FRAME 상부로 올라가야 합니다. 이에 따라 추락 위험이 발생할 수 있어, 안전성 확보를 위해 해당 스위치를 나인벨 로봇의 인터페이스 패널로 이동하고자 합니다. In order to operate the TP-HOST key switch installed on the control box, the operator must climb onto the upper part of the frame. This may pose a risk of falling. Therefore, to improve safety, the switch will be relocated to the interface panel of the NINEBELL robot.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	TP/HOST 스위치는 컨트롤러 박스에 위치한다. The TP/HOST key switch is currently installed on the controller box.	SI5-C00L1-1(Controller Panel), TI1-Z00P5-0(INTERFACE BKT), AN-Z00L1-0(Z AXIS INTERFACE PANNEL) 품목과 동일한 역할의 가공품을 수정하여 스위치의 위치를 컨트롤러 박스에서 나인벨의 로봇 판넬로 이동한다. By modifying the machined parts that perform the same function as SI5-C00L1-1 (Controller Panel), TI1-Z00P5-0 (Interface Bracket), and AN-Z00L1-0 (Z-Axis Interface Panel), the TP/HOST key switch will be relocated from the controller box to the NINEBELL robot panel.	

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전체 이미지 Full Image	    ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions					
유효성 검토 결과 Test Result	검토 여부 Test or not	Y / N	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approval	Y / N
승인 날짜 Approval date	1. Ninebell : 2026.03.06 2. Customer : / /		나인벨 승인 Ninebell approval	QC 2026.03.06 PASS	고객사 승인 Customer approval	ACM confirm 3/10 Zhaoming Zhong
적용 일자 Effective date	After Approval 2026.03.10		최초 적용 장비 Initially Applied Equipment	After Approval HP87-007		